Probe Measurements Force Finite Centers Only Through Center-Separating Effects

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Abstract

Local probe-measurement frameworks in algebraic quantum field theory provide a causal and covariant way to define induced system observables and instruments. They are therefore a possible positive route out of finite-center nonidentifiability. This note isolates the finite-central boundary. A probe/instrument family reconstructs a finite central target exactly when the induced system effects separate the corresponding central contrast fibers. If all induced effects factor through a quotient, retraction, or coarse-graining, then finite adaptive composition preserves that blindness. Thus a measurement formalism alone does not force hidden finite central rank or the predicate $q = 2$; the supplied probe effects must be informationally complete for that target, or a physical quotient/exclusion axiom must remove the hidden center. Conversely, induced effects proportional to the minimal central projections give a direct positive exit: they reconstruct the retained finite central state and separate every binary-vs-nonbinary contrast in that retained center.

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1 Primary sources

The boundary below was checked against the Fewster–Verch local measurement framework [1], the Fewster–Verch review of measurement in QFT [2], the LCQFT/dynamical-locality setting [3], and recent work on measurements in the Fewster–Verch framework [4]. These works are used here as positive measurement constructions, not as claims that finite-center separation is absent from physics.

2 Finite central readout

Let $Z = \ell^\infty(\{1, \ldots, q\})$ be a finite center. Its real contrast space is

$$V_q = \left\{ x = (x_1, \ldots, x_q) \in \mathbb{R}^q : \sum_i x_i = 0 \right\}.$$

A declared family of probe preparations, couplings, and probe readouts induces system effects. Restricting those induced effects to Z gives vectors $e_\alpha \in \mathbb{R}^q$, hence central contrast functionals

$$x \longmapsto \langle e_\alpha, x \rangle.$$

Let

$$N = \{x \in V_q : \langle e_\alpha, x \rangle = 0 \text{ for every supplied induced effect } \alpha\}.$$

Theorem 1 (probe effect-span criterion)**.** *Let $T : V_q \to X$ be a finite central target. Then T is reconstructible from the supplied probe-measurement data if and only if T is constant on cosets of N:*

$$x - y \in N \quad \Longrightarrow \quad T(x) = T(y). \tag{1}$$

In particular, the full finite central contrast is reconstructible exactly when

$$\operatorname{span}\{e_\alpha - \bar{e}_\alpha \mathbf{1}\}_\alpha = V_q^*, \qquad \bar{e}_\alpha = \frac{1}{q} \sum_i (e_\alpha)_i. \tag{2}$$

Proof. The supplied central measurement record is the linear map

$$R : V_q \to \mathbb{R}^{\mathcal{I}}, \qquad R(x)_\alpha = \langle e_\alpha, x \rangle,$$

whose kernel is N. A target is a function of the measurement record exactly when it is constant on the fibers of R, which is (1). Full contrast reconstruction means that R is injective on V_q, equivalently that the restrictions of the induced effects span V_q^*, which is (2). □

Proposition 1 (atomic induced effects are centrally complete)**.** *Assume the supplied probe-measurement family contains induced system effects whose restrictions to $Z = \ell^\infty(\{1, \ldots, q\})$ are the minimal central projections $e_1, \ldots, e_q$, or any invertible linear recombination of them modulo the scalar unit. Then the protocol reconstructs the retained finite central state on Z. In particular, it separates the binary predicate $q = 2$ from every retained nonbinary comparison model once the retained center itself is known to be the whole finite central target.*

Proof. For a central state $p = (p_1, \ldots, p_q)$, the probability of the induced effect e_i is p_i. Thus the atomic effects recover the full probability vector p. Equivalently, on the contrast space V_q, the classes of e_i modulo the scalar unit span V_q^*, so Theorem 1 applies. An invertible linear recombination gives the same span. The final binary-rank statement uses the additional premise that the measured center is the full finite central target; otherwise one has reconstructed only the retained quotient center. □

3 Adaptive protocols

Proposition 2 (adaptivity does not create a missing central separator)**.** *Suppose each stage of a finite adaptive protocol has induced central effects that factor through a fixed central coarse-graining*

$$C : \{1, \dots, q\} \to \{1, \dots, r\}.$$

Then every joint outcome probability of the finite protocol factors through the same coarse-graining. Consequently no target that varies inside a C-fiber is reconstructible from the protocol.

Proof. At each stage, conditioning on previous outcomes may choose a different coupling, probe state, or probe effect, but by hypothesis the induced central functional is constant on the fibers of C. Products, sums, and conditional branches of functions constant on C-fibers remain constant on C-fibers. Induction over the finitely many stages shows that every joint outcome probability is constant on C-fibers. Any target that changes within such a fiber therefore fails the fiber-constancy condition of Theorem 1. □

4 Positive exits and referee question

Exit premise	Effect
Center-separating induced effects	The probe observables span the central contrast dual V_q^*, so finite central states are tomographically visible.
Direct atomic central readout	Effects proportional to the minimal central projections e_i reconstruct the retained central state by Proposition 1.
Factor, quotient, or inadmissibility axiom	The hidden finite center is removed from the comparison class rather than reconstructed by measurement.
Full physical tomography assumption	If the theorem assumes tomography for the full physical algebra, finite central separation is part of the hypothesis.

The exact external question is:

> Does the local probe-measurement setup in the target setting supply induced system effects that separate the finite central contrasts, or does it only provide a causal measurement framework whose chosen effects may still factor through a quotient or retraction?

If the first holds, the measurement scheme is a positive exit from the Team 1 finite-center obstruction. If the second holds, the framework alone does not force hidden finite central rank or $q = 2$.

References

[1] C. J. Fewster and R. Verch, *Quantum fields and local measurements*, Commun. Math. Phys. 378 (2020), 851–889, arXiv:1810.06512.

[2] C. J. Fewster and R. Verch, *Measurement in Quantum Field Theory*, arXiv:2304.13356.

[3] C. J. Fewster and R. Verch, *Dynamical locality and covariance: What makes a physical theory the same in all spacetimes?*, Ann. Henri Poincare 13 (2012), 1613–1674, arXiv:1106.4785.

[4] J. Mandrysch and M. Navascués, *Quantum Field Measurements in the Fewster–Verch Framework*, arXiv:2411.13605.